

OMAR MAHMOUD

DATA ENGINEER | MACHINE LEARNING ENGINEER

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFILE

I am a Computer Science student at Helwan University, specializing in Data Engineering through the Digital Egypt Pioneers Initiative (DEPI). I have strong expertise in designing and implementing ETL/ELT pipelines, data preprocessing, SQL optimization, and data visualization using Power BI and Tableau. I am proficient in Python, SQL, Apache Spark, Apache Airflow, and Apache Kafka, with hands-on experience in developing and managing data workflows and machine learning pipelines for predictive analytics and data-driven solutions. Throughout my academic and personal projects, I have gained practical experience in data engineering, machine learning, and full-stack development. I am highly motivated to transform raw data into scalable, efficient systems that generate meaningful and actionable business insights.

EDUCATION

Bachelor of Computer Science at Helwan University, Cairo, Egypt **August 2024 - July 2028**
Grade: Very Good - 3.32/4 (Degree in Progress)

INTERSHIPS

Data Engineer Trainee at [Digital Egyptian Pioneers initiative](#), Cairo - Egypt **November 2025 - July 2026**

- Designed and maintained ETL/ELT pipelines for analytics and enterprise reporting.
- Cleaned, transformed, and integrated data using Python (Pandas) and advanced SQL.
- Optimized relational database schemas to improve query performance.
- Applied data modeling, orchestration, and data quality best practices.

CERTIFICATES

Data Engineering - AI & Data Science from Digital Egypt Pioneers Initiative • [Link](#) **July 2026**
Completed a professional certification through the Digital Egypt Pioneers Initiative (DEPI), focusing on Data Engineering, AI, and Data Science fundamentals. The program covered data processing, ETL pipelines, data analysis, and machine learning concepts using industry-relevant tools and technologies.

Machine Learning from National Telecommunication Institute • [Link](#) **May 2026**
Completed a professional certification in Machine Learning from the National Telecommunication Institute (NTI), focusing on core ML concepts, supervised and unsupervised learning, model training, and evaluation techniques.

PROJECTS

Customer-Churn-Prediction-Pipeline • [GitHub](#) **December 2025 - May 2026**
Tech Stack: Python, Pandas, Scikit-learn, SQL, Streamlit, Power BI
Developed an end-to-end machine learning pipeline to predict customer churn using a dataset of 999K+ customer

records and 32 features. Performed data preprocessing, feature engineering, exploratory data analysis, and model evaluation using Python and SQL. Developed a Logistic Regression model achieving 0.6815 ROC-AUC and identifying 63% of churn-risk customers. Designed an interactive Streamlit dashboard for churn analytics, KPI monitoring, and customer risk prediction.

Monitor Home System • [GitHub](#)

March 2026 - May 2026

Developed a full-stack home monitoring web application using PHP, MySQL, HTML, CSS, and JavaScript. Designed backend services for managing and processing home system data efficiently. Implemented responsive user interfaces and core CRUD functionalities for real-time monitoring.

Hyper-Market-System • [GitHub](#)

November 2025 - January 2026

Built a Java-based console application using OOP principles to manage products, stock, and categories. Implemented inventory tracking and product management functionalities to improve operational organization.

Helwan-Wings • [GitHub](#)

November 2025 - January 2026

Designed and developed an airport management and ticket booking system with database integration. Created system analysis and design diagrams including UML diagrams and database schema. Implemented functionalities for flight scheduling, passenger management, and ticket booking.

Fashion-shop • [GitHub](#)

April 2025 - May 2026

Developed a responsive front-end e-commerce website using HTML, CSS, and JavaScript. Implemented product filtering, shopping cart management, and product detail pages. Improved user experience through responsive layouts and modern UI design practices.

SKILLS

Programming: Python (Pandas, NumPy, Matplotlib), SQL, Java (Basic), Data Structures & Algorithms

Databases & Data Storages: Relational Databases (MySQL, PostgreSQL), Database Design, NoSQL (MongoDB)

Data Engineering & Big Data Tools: Apache Spark, Apache Airflow, Apache Kafka, Hadoop, ETL/ELT Pipelines, Azure Data Factory

Data Visualization & Tools: Power BI, Tableau, Excel (Advanced), Jupyter Notebook, Visual Studio Code

Cloud & DevOps Tools: Microsoft Azure / AWS (Fundamentals), Docker, Kubernetes (Basic), CI/CD Concepts, Git & GitHub

LANGUAGES

Arabic - Native

English - Proficient